



FEA REPORT

Static Structural Analysis Of Guillotine Gate & Frame

Report No.: MWI/FEA/ACK-6179/01

Date: 4th Sept 2026

Client: SILVERTONE





1. Material Properties

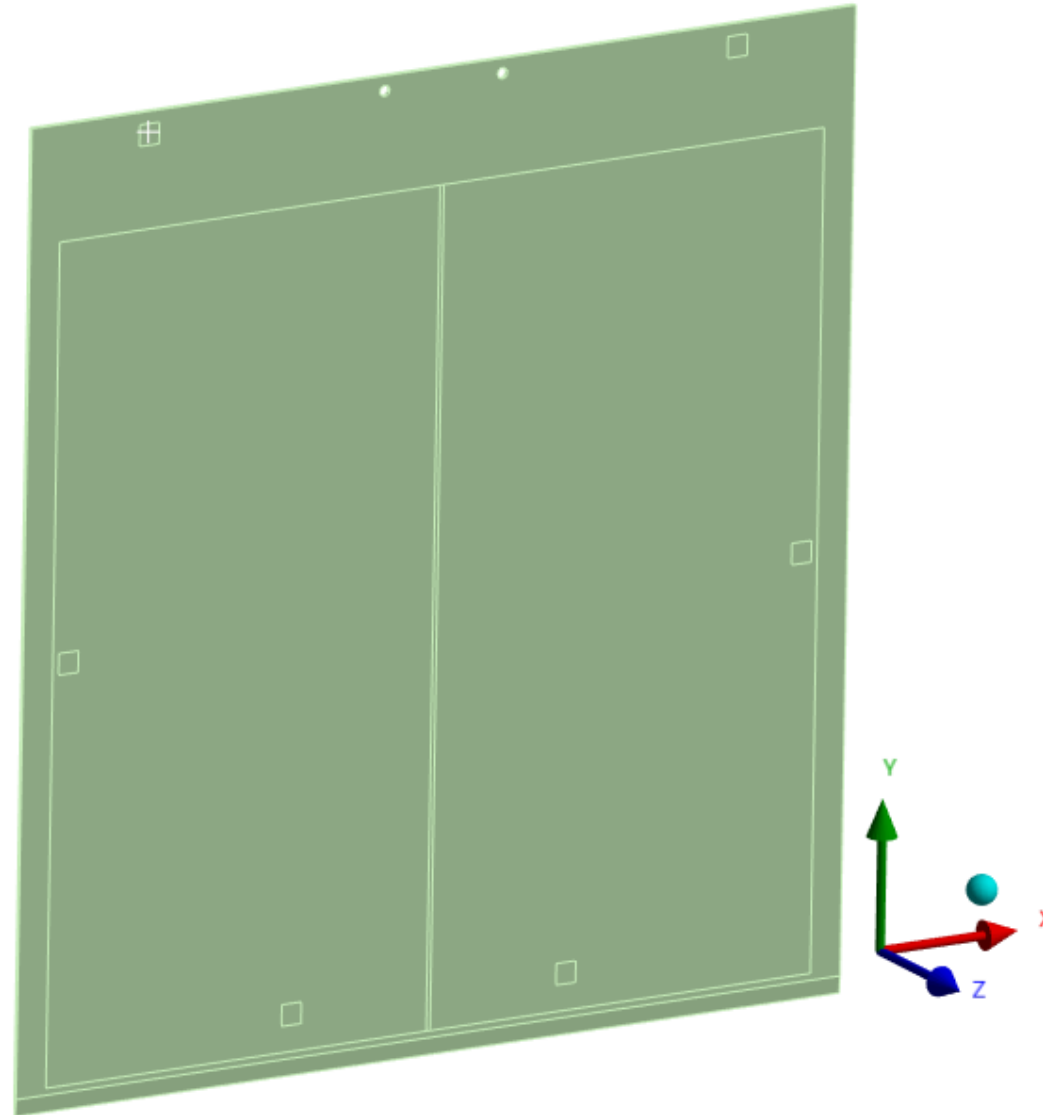
Material	Item	Young's Modulus (MPa)	Poisson's Ratio	Density (t/mm ³)	Yield Stress (MPa)	Allowable Von-Mises Stress (MPa)
IS 2062	Duct	191400	0.3	7.8E-9	213	163

Units: Length = mm, Mass = tonnes, Time = second, Force = Newton

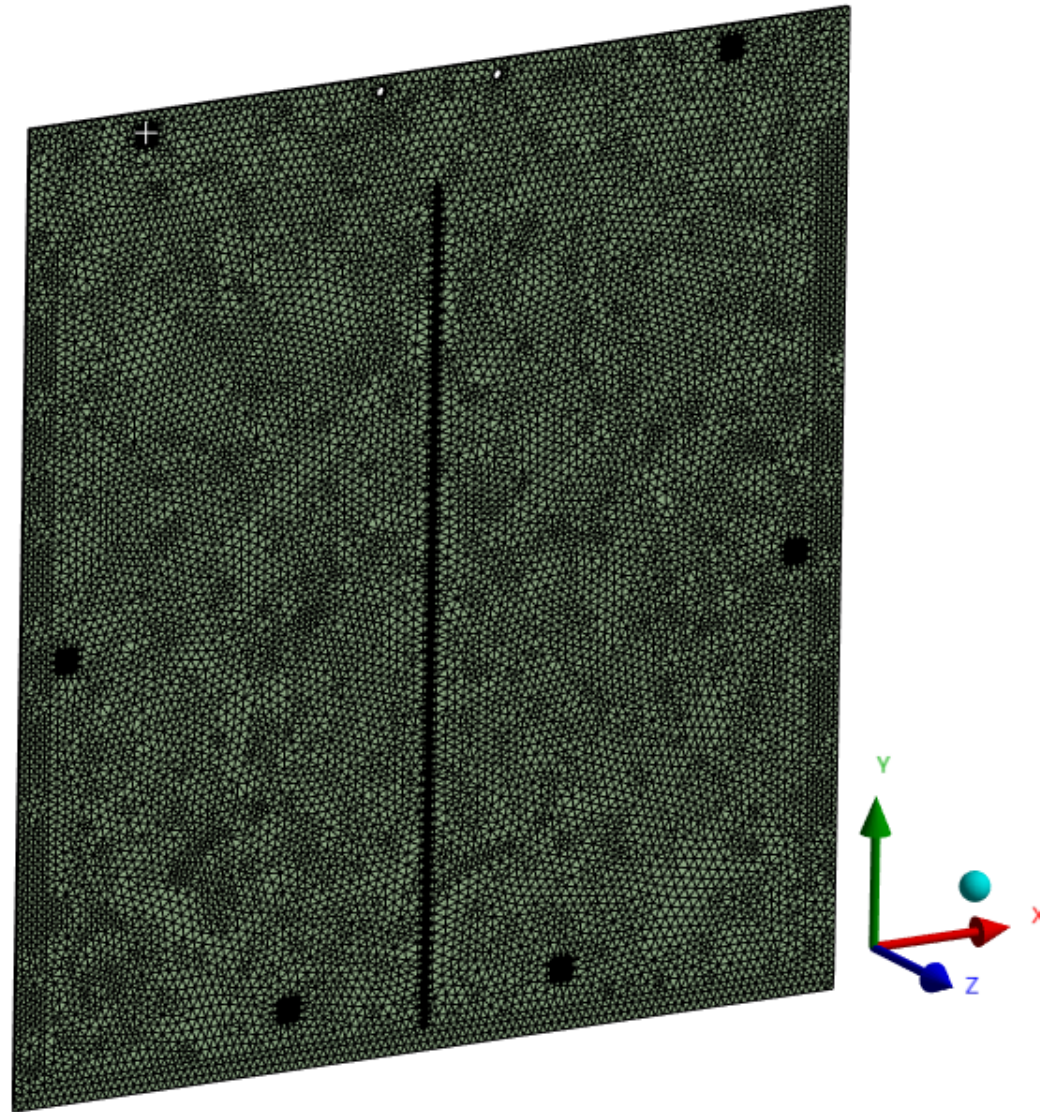
Allowable Von-Mises Stress:

$\sigma_{all} = S_{yt} / FOS$ **FOS = 1.3**

2. GG Geometry



3. GG Mesh



Case 1: Guillotine Gate Fully Closed (Self Weight + Pressure)



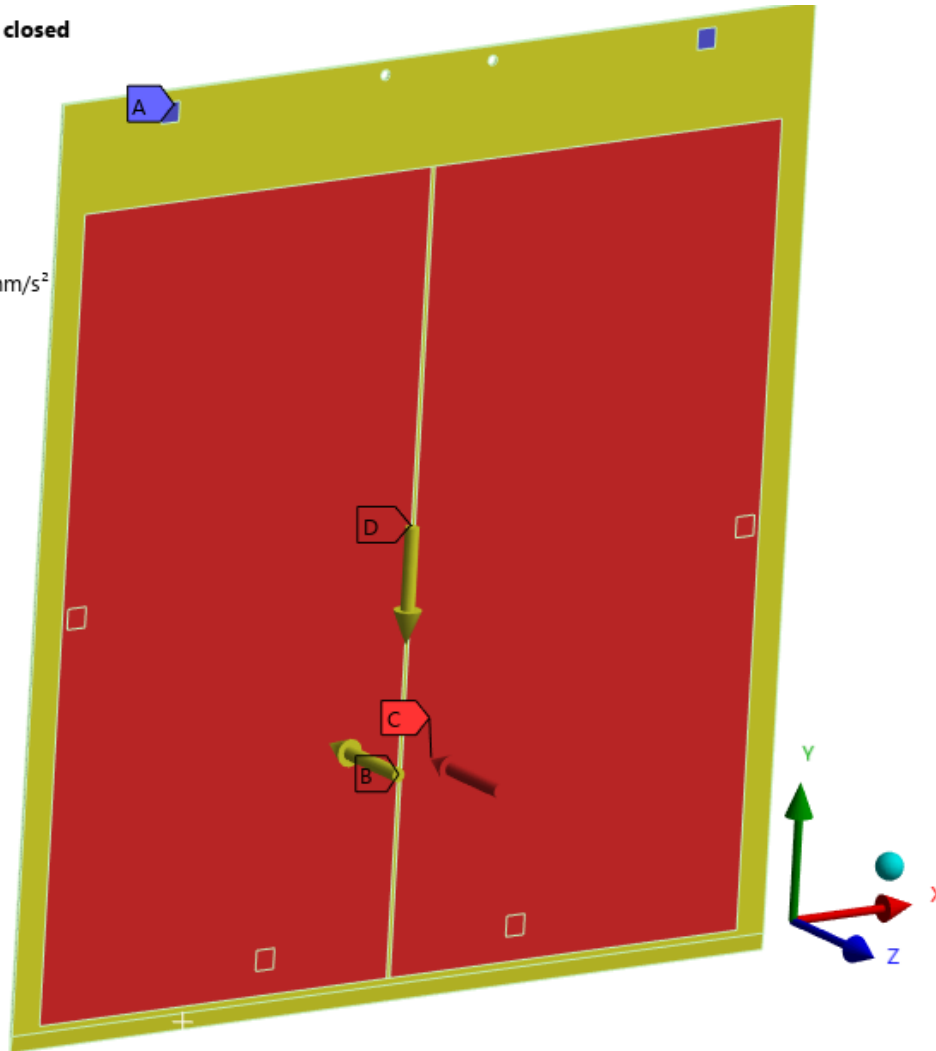
A: Static Structural Silverstone GG closed

Static Structural

Time: 0.99723 s

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- A** Fixed Support
- B** Displacement
- C** Pressure: 6.09e-003 MPa
- D** Standard Earth Gravity: 9806.6 mm/s²



Boundary Conditions

- A. Roller contact area is been Fixed.
- B. Free Displacements of 2 mm in Z Direction.
- C. Design pressure 621.47mmWC applied which is 0.00609 MPa.
- D. self weight considered.

Note: 1. This load case is at the working temperature of 210°C.
2. Boundary conditions were applied to match the physical constraints of the geometry.

Total Deformation

Total deformation

A: Static Structural Silverstone GG closed

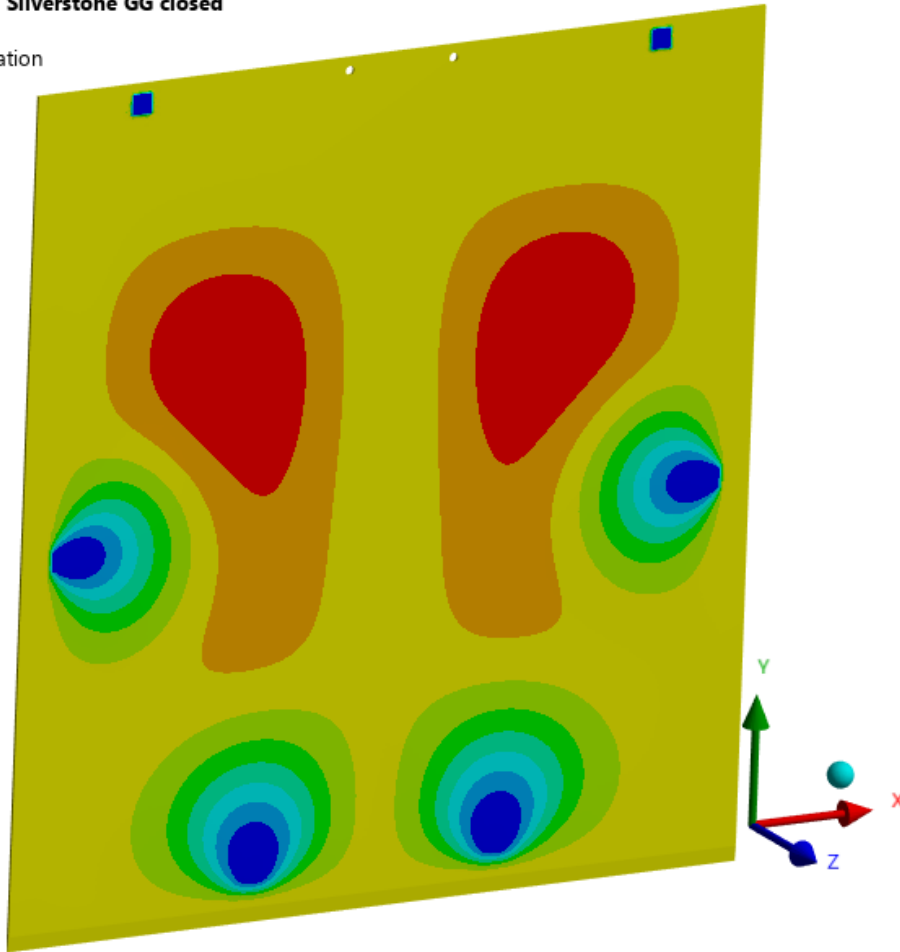
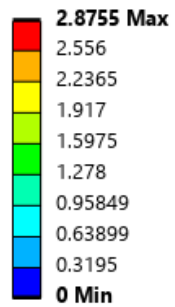
Total Deformation

Type: Total Deformation

Unit: mm

Time: 1 s

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Maximum Total Deformation 3 mm

Von-mises stress

A: Static Structural Silverstone GG closed

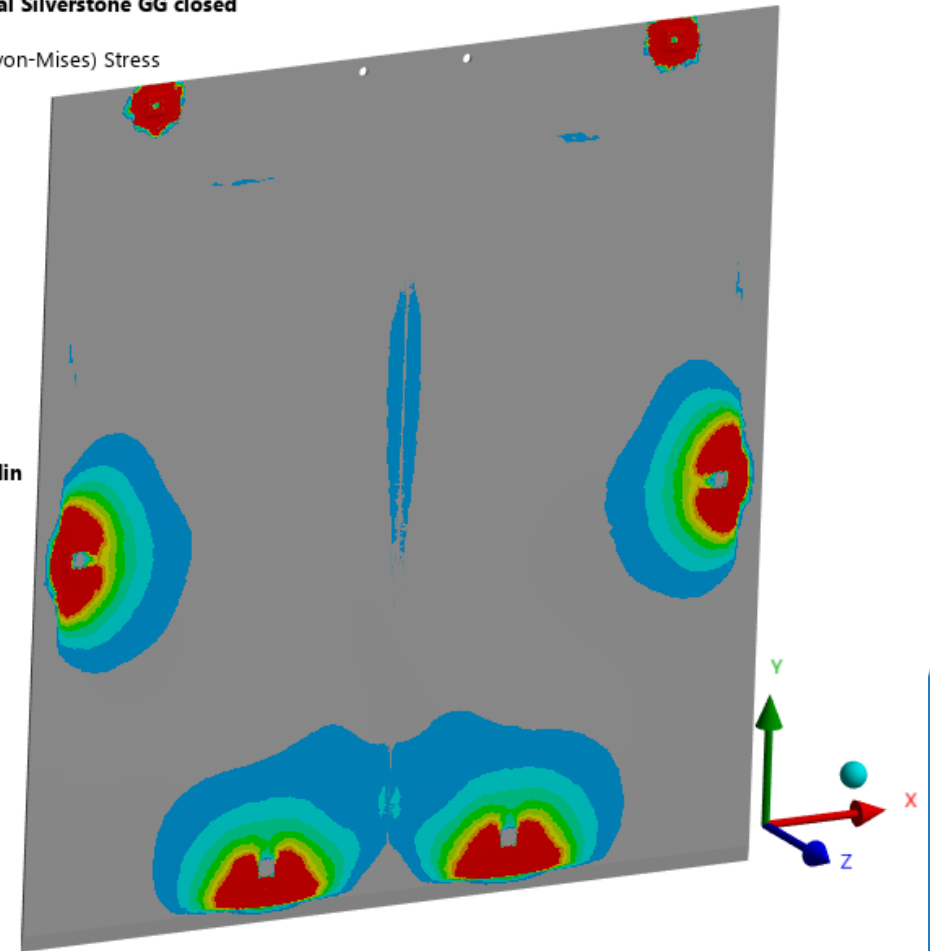
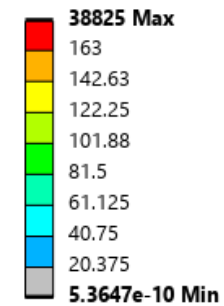
Equivalent Stress

Type: Equivalent (von-Mises) Stress

Unit: MPa

Time: 1 s

04-09-2026 13:35



The stress exceeds the materials allowable limit of 163 MPa.



Observation & Summary

- Total deformation in the GG is 3 mm.
- Von-Mises stress in the GG exceeds the materials allowable stress limit of 163 MPa.
- Therefore, the GG is structurally inadequate to withstand the applied load under this load case.
Need to take corrective actions.



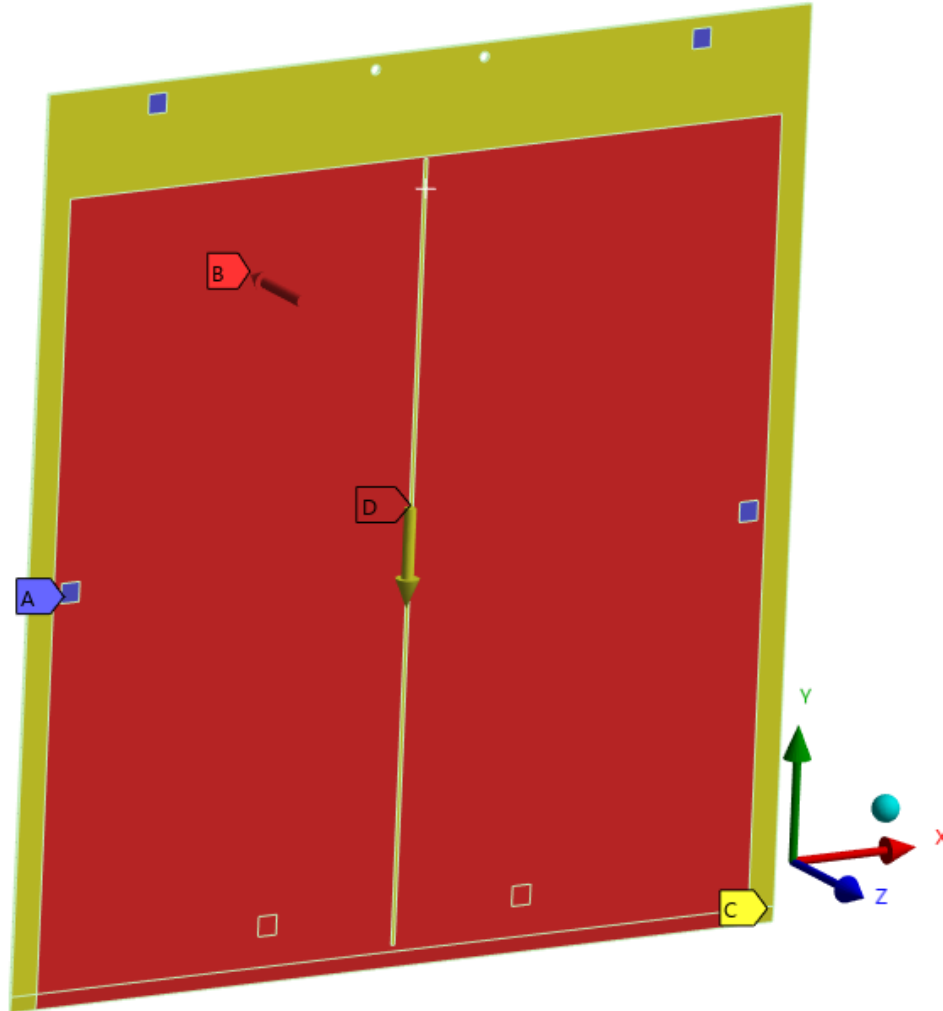
Case 2: Guillotine Gate Partially Closed (Self Weight + Pressure)

B: Static Structural Silverstone guillotine gate partially closed

Static Structural

Time: 0.99723 s

- A** Fixed Support
- B** Pressure: 6.09e-003 MPa
- C** Displacement 2
- D** Standard Earth Gravity: 9806.6 mm/s²



Boundary Conditions

- A. Roller contact area is been Fixed.
- B. Design pressure 621.47mmWC applied which is 0.00609 MPa.
- C. Free Displacements of 2 mm in Z Direction.
- D. Self weight considered

Results



Total deformation

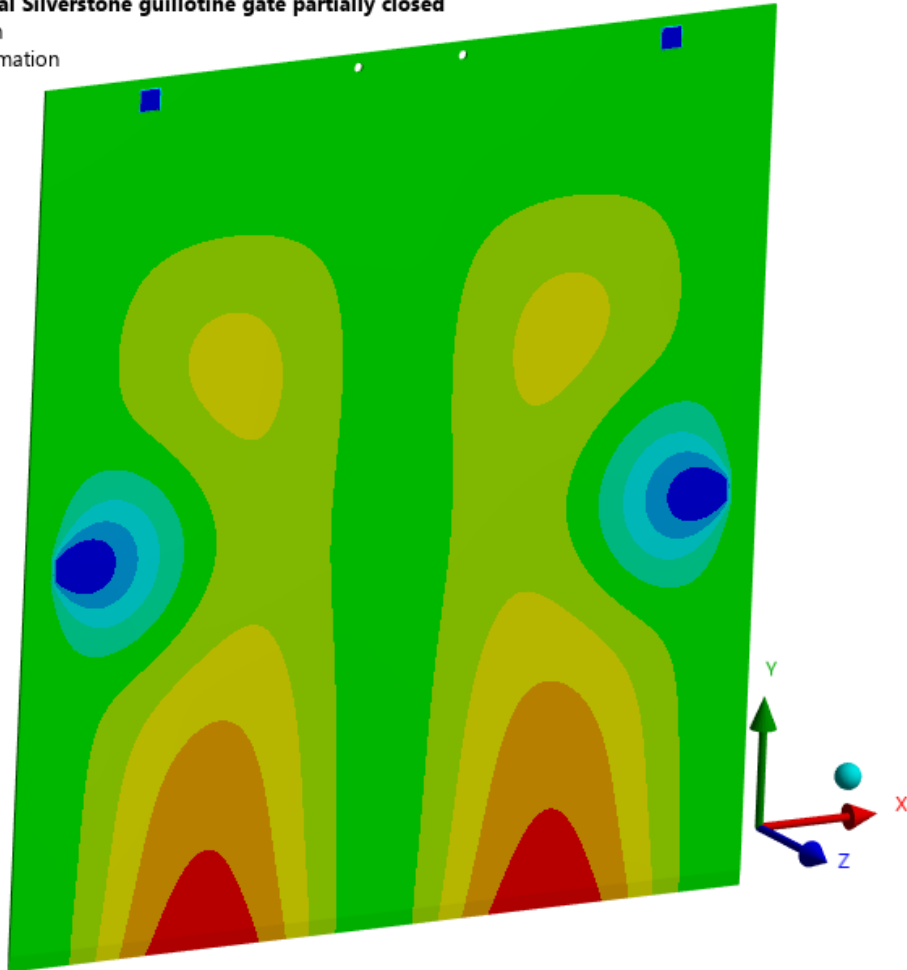
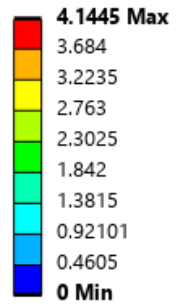
B: Static Structural Silverstone guillotine gate partially closed

Total Deformation

Type: Total Deformation

Unit: mm

Time: 1 s



Maximum Total Deformation – 4 mm

Von-mises stress

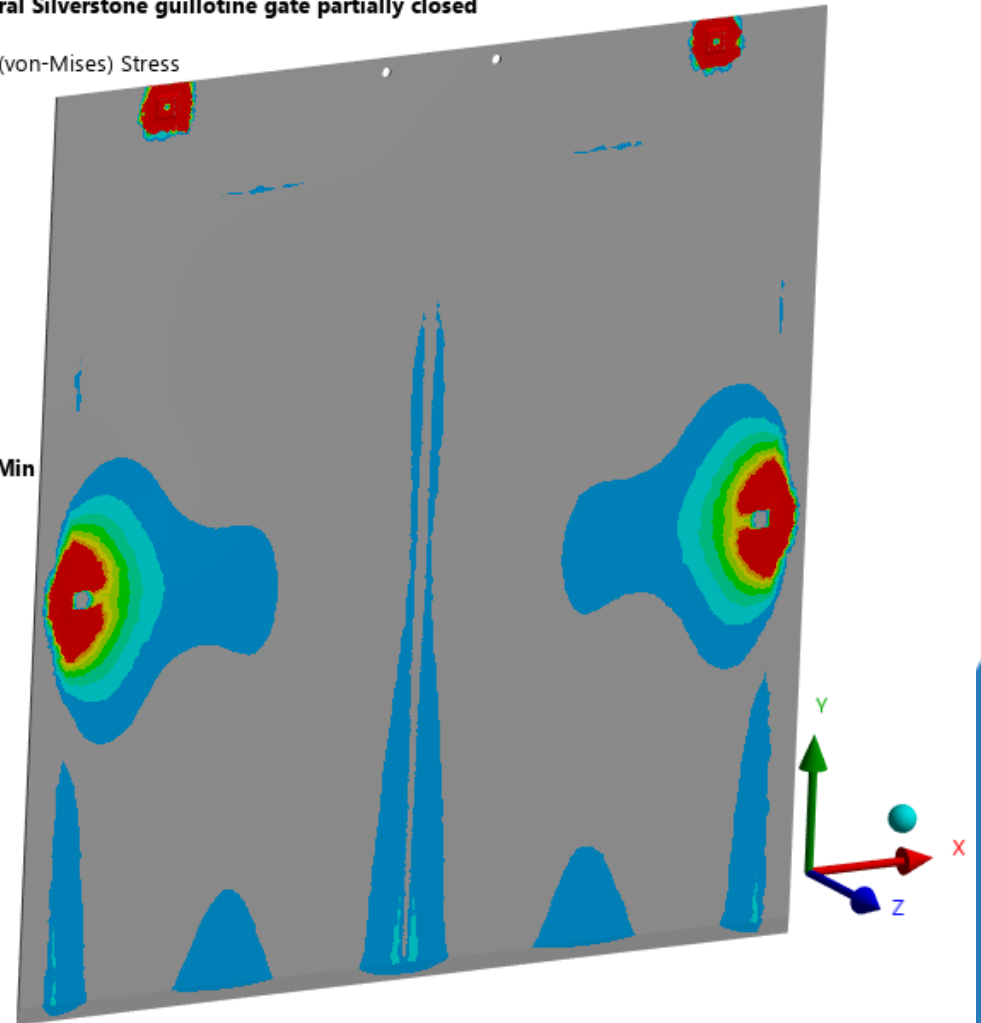
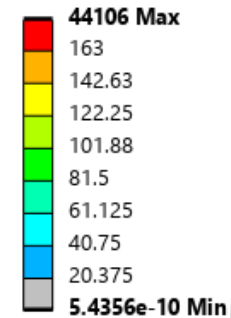
B: Static Structural Silverstone guillotine gate partially closed

Equivalent Stress

Type: Equivalent (von-Mises) Stress

Unit: MPa

Time: 1 s



The stress exceeds the materials allowable limit of 163 MPa.



Case 2: Observation & Summary

- Total deformation in the GG is 4 mm.
- Von-Mises stress in the GG exceeds the materials allowable stress limit of 163 MPa.
- Therefore, the GG is structurally inadequate to withstand the applied load under this load case.
Need to take corrective actions.

Final Summary



Load Case	Max. Deformation	Max. Von-Mises Stress	Allowable Stress
Case – GG Fully closed	3 mm	Exceeding Limit	163 MPa
Case – GG Partially closed	4 mm	Exceeding Limit	163 MPa